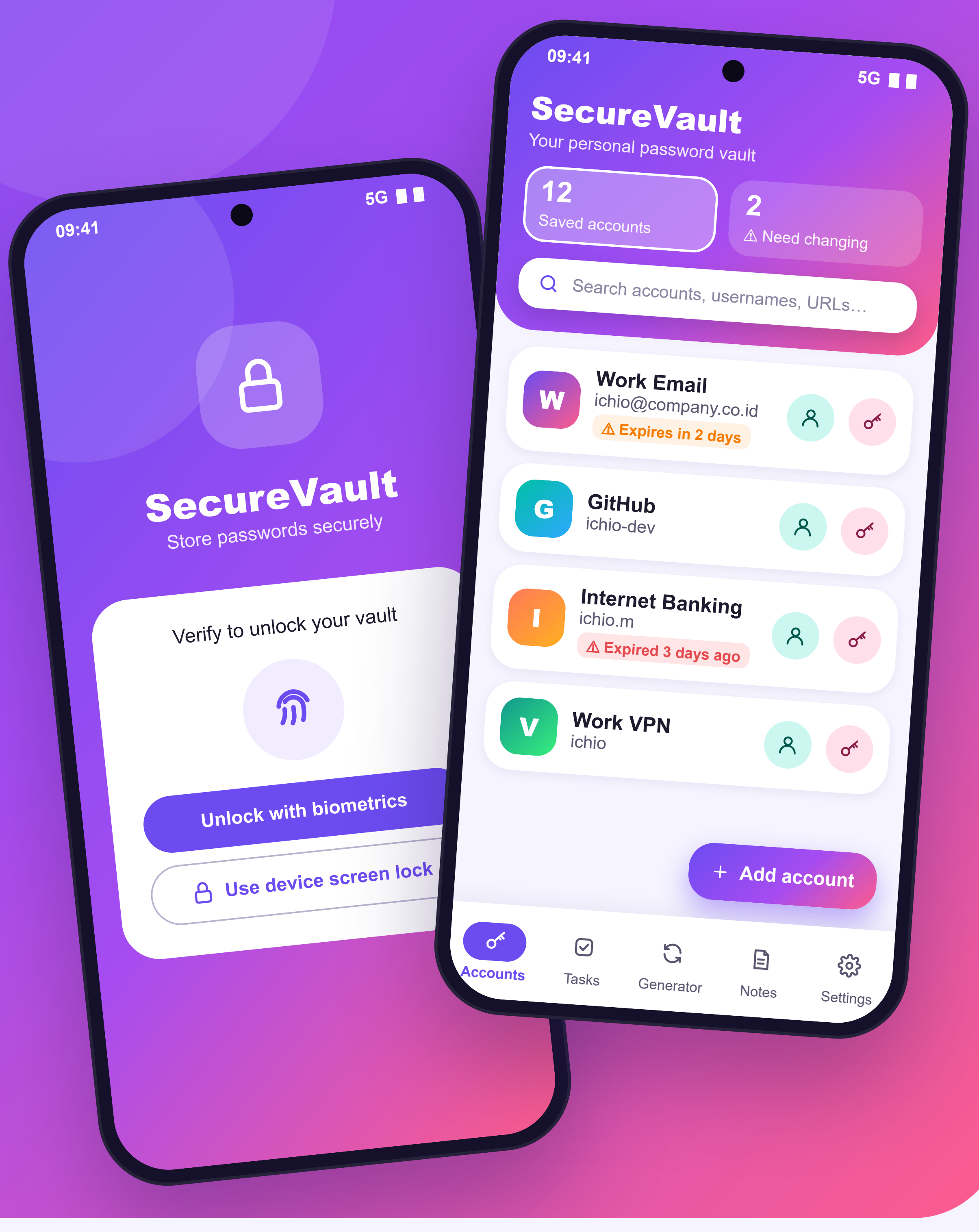
SecureVault · v1.4

A secure, smart & modern personal password vault.

Store accounts, create strong passwords, check for data breaches, and get reminders — all encrypted on your phone.

- Anti-screenshot
- AES-256-GCM
- HIBP breach check
- Expiry reminders
- Biometric / device lock
- Android 8.0+

Native Android app (Kotlin + Jetpack Compose) · Built by **Ichio**



09:41 5G

SecureVault
Store passwords securely

Verify to unlock your vault

09:41 5G

SecureVault
Your personal password vault

12 Saved accounts
2 ⚠ Need changing

Work Email
ichio@company.co.id
⚠ Expires in 2 days

GitHub
ichio-dev

Internet Banking
ichio.m
⚠ Expired 3 days ago

Work VPN
ichio

KEY STRENGTHS

6 reasons your data is safer in SecureVault

01 Can't be captured

Screenshots & screen recording are blocked by the system (FLAG_SECURE). App content is also hidden in the Recent Apps list.

✓ Screen always protected

02 Data-breach check

Integrated with Have I Been Pwned: check a password, audit all stored passwords, and check whether an email has been breached.

Passwords are never sent

03 Expiring-password reminders

Set a validity of 30–365 days. Automatic notification 2 days before expiry, plus orange/red markers in the account list.

Checked every 6 hours in the background

04 AES-256 encryption

Passwords & notes are encrypted with AES-256-GCM using an Android Keystore key, and the whole database is encrypted with SQLCipher — keys never leave the device.

AES-256-GCM · SQLCipher

05 Biometric / device lock

Unlock with a strong biometric or your device screen lock. Unlocking releases the auth-bound vault key — a cryptographic lock, not just a UI gate. Auto-locks 30 seconds in the background.

Auto-lock in 30 s

06 Encrypted backup & restore

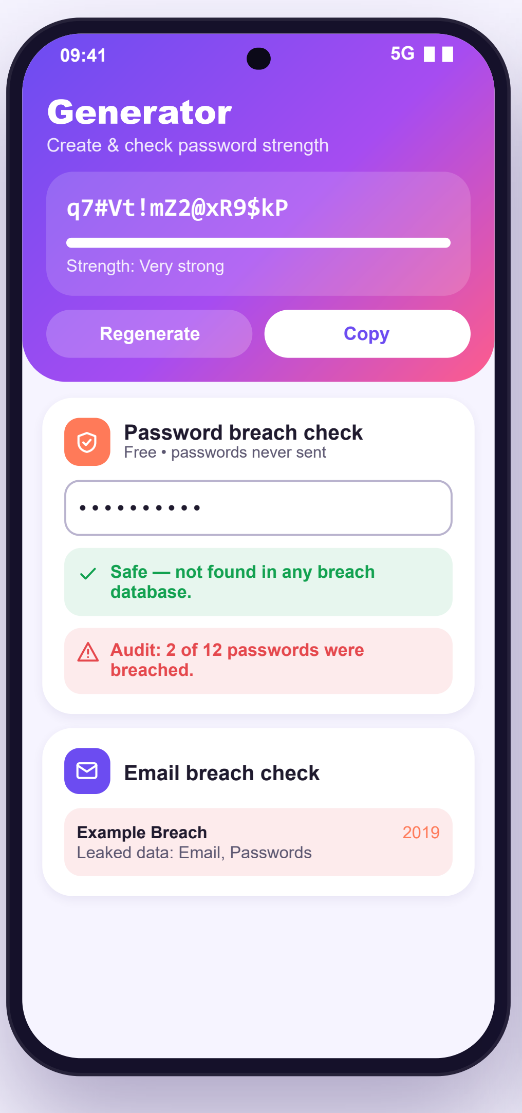
All accounts, notes & tasks in a single password-locked .svbk file (PBKDF2, 210,000 iterations). Easy to move to a new phone.

Merge / replace all

APP SCREENS

Modern, bright, and easy to use

Screen illustrations are recreated from the app's design — real screenshots can't be taken because SecureVault blocks screen capture.



09:41 5G

Generator
Create & check password strength

`q7#Vt!mZ2@xR9$kP`

Strength: Very strong

Password breach check
Free • passwords never sent

••••••••••

✓ Safe — not found in any breach database.

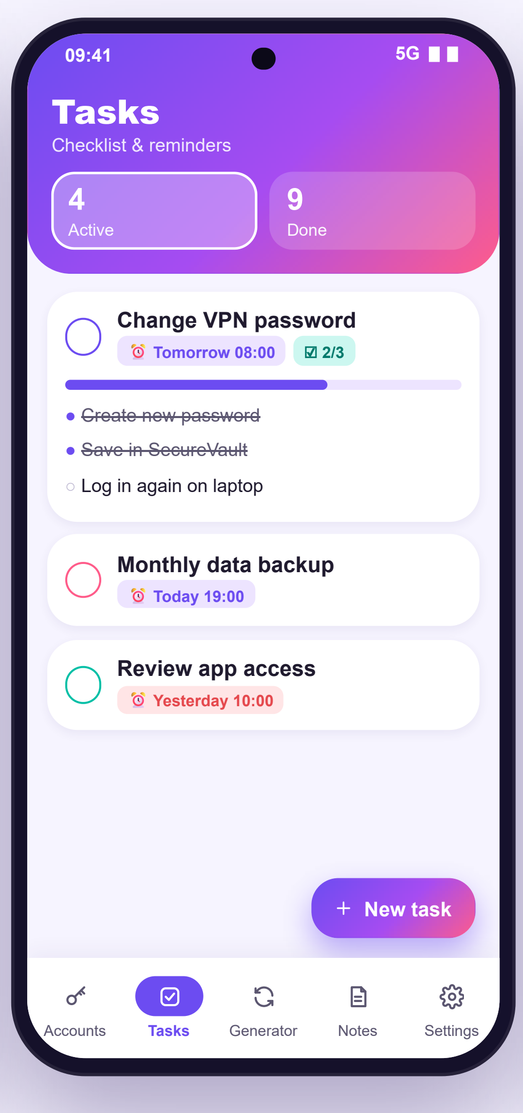
⚠ Audit: 2 of 12 passwords were breached.

Email breach check

Example Breach 2019
Leaked data: Email, Passwords

Generator & breach check

Strong random passwords plus a Have I Been Pwned check on one screen.



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Tasks
Checklist & reminders

4 Active
9 Done

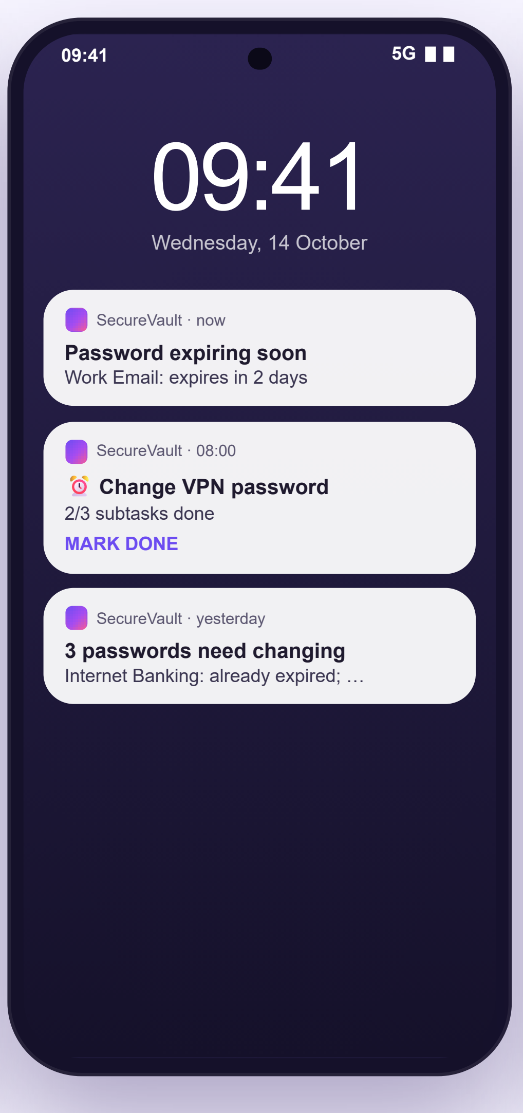
○ Change VPN password
Tomorrow 08:00 · ☑ 2/3
- ~~Create new password~~
- ~~Save in SecureVault~~
- Log in again on laptop

○ Monthly data backup
Today 19:00

○ Review app access
Yesterday 10:00

Tasks & subtasks

Checklists with progress, auto-completed when all subtasks are ticked.



09:41 5G

09:41
Wednesday, 14 October

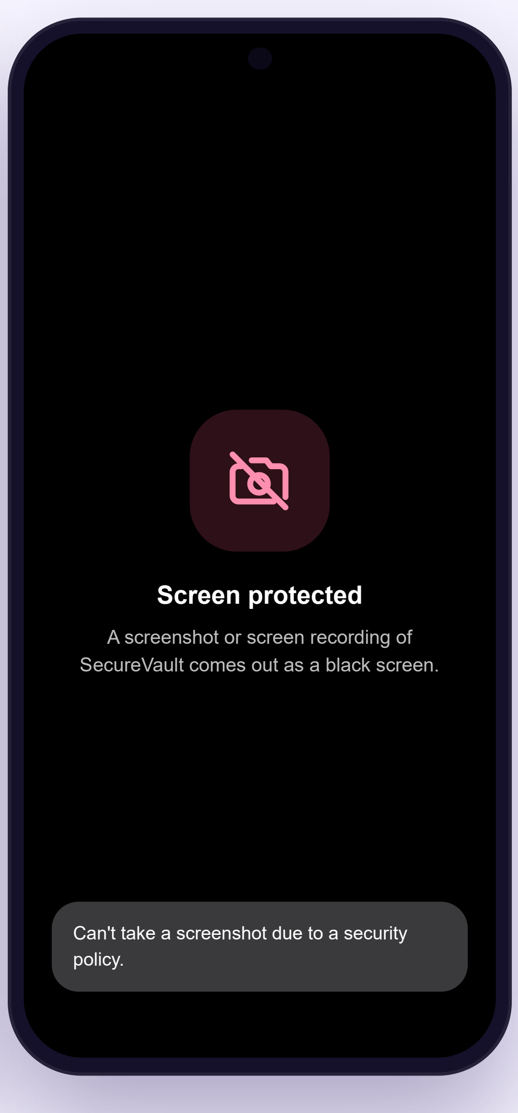
SecureVault · now
Password expiring soon
Work Email: expires in 2 days

SecureVault · 08:00
Change VPN password
2/3 subtasks done
MARK DONE

SecureVault · yesterday
3 passwords need changing
Internet Banking: already expired; …

Smart notifications

Expiry reminders (2 days ahead) & task reminders — without showing passwords.



Screen protected

A screenshot or screen recording of SecureVault comes out as a black screen.

Can't take a screenshot due to a security policy.

Anti-capture

Protects passwords from screenshots, screen recording & screen-sharing apps.

HOW THE BREACH CHECK WORKS

Passwords are checked without ever being sent

SecureVault uses the **k-Anonymity** method against the Pwned Passwords database from Have I Been Pwned. The final match happens on your phone.

1. **Password on phone**
   The password you want to check stays on the device.
   `P@ssw0rd123`
2. **SHA-1 hash** →
   Turned into an irreversible 40-character hash.
   `0F0D9…E7C5F`
3. **Send 5 characters** →
   Only the first 5 characters of the hash are sent to the HIBP API.
   `/range/0F0D9`
4. **Match locally** →
   HIBP returns hundreds of similar hashes; the phone matches them itself.
   `match? → breach count`

The HIBP server never learns your password. Email checks use an HIBP API key that is stored encrypted.

PASSWORD EXPIRY LIFECYCLE

No more passwords you forget to change

Save account
Choose "Change every 90 days" or a specific date
Day 0

Monitored automatically
Background check every 6 hours
Every day

2-day notice
"Password expires in 2 days"
Day 88

Expired
Marked red & shown in the "Needs changing" filter
Day 90

Change password
Validity is automatically recalculated to 90 days
New cycle

SECURITY LAYERS

Layered protection

- **App lock**
  Biometric / device credential · auth-bound key · auto-lock
- **Protected screen**
  Blocks screenshots, screen recording & Recent Apps preview
- **Safe clipboard**
  Marked sensitive · auto-cleared after 30 seconds
- **Data encryption**
  AES-256-GCM · SQLCipher database · Android Keystore
- **Local storage**
  No cloud · Android backup disabled · no ads/trackers

FULL FEATURE SET

Everything in one app

Password manager Usernames, passwords, URLs, notes & search	**Generator** 6–64 characters, SecureRandom, strength indicator
Breach check Passwords, bulk audit & email (HIBP)	**Expiry** Interval / date, 2-day notice
Tasks & subtasks Checklist, reminders, "Mark done"	**Secure notes** Encrypted note content, colorful grid view
Backup & restore Encrypted .svbk file, move between phones	**Biometric unlock** Fingerprint / face / device screen lock
Offline-first Internet only when a breach check is requested	**Full help** A 13-topic guide right in the app

- Android 8.0 (API 26) and up
- Kotlin · Jetpack Compose · Material 3
- Room + SQLCipher
- WorkManager & AlarmManager
- PBKDF2-SHA256 · AES-256-GCM · SHA-1 k-anon
- Have I Been Pwned API v3